EE 121B: Principles of Semiconductor Device Design

Lecture 2:

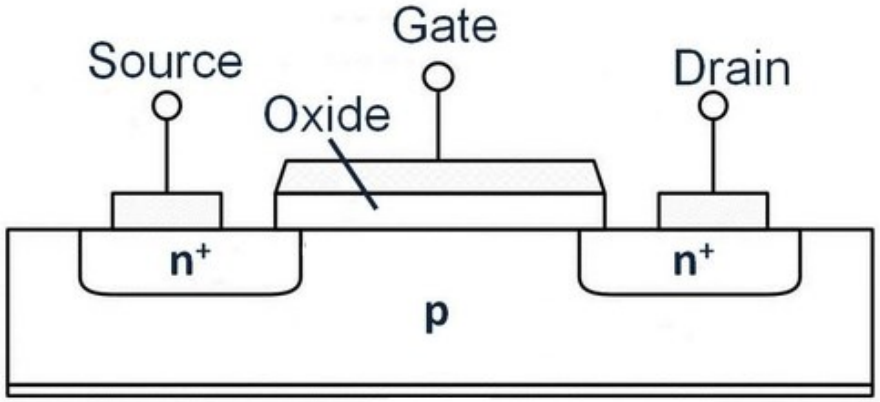
Semiconductor Fundamentals

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. P-N Junction
3. Metal-Semiconductor Contacts
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

Lecture 2

OUTLINE

- Semiconductor Fundamentals (cont'd)
  - Bond model
  - Energy band model
  - Band gap energy
  - Density of states
  - Doping

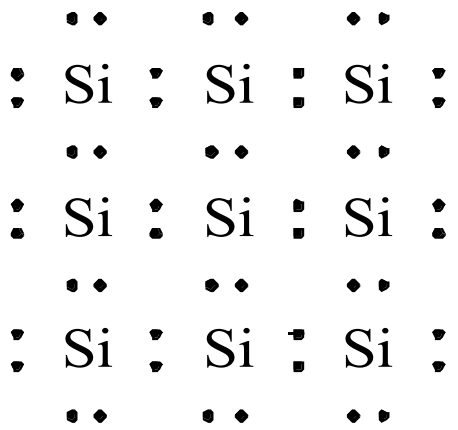
Reading: Pierret 2.2-2.3, 3.1.5; Hu 1.3-1.4,1.6, 2.4

Recall

- Crystalline Si:
  - 4 valence electrons per atom
  - diamond lattice (each atom has 4 nearest neighbors)
  - **atomic density = 5 x 10^{22} atoms/cm^3**
  - **intrinsic carrier concentration n_i = 10^{10} cm^{-3}**
  - Miller indices are used to designate planes and directions within a crystalline lattice
- In a pure Si crystal, conduction electrons and holes are formed in pairs.
  - Holes can be considered as positively charged mobile particles.
  - Both holes and electrons can conduct current.

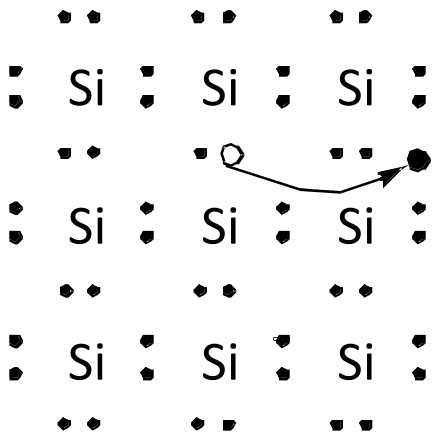
Recall: Electrons and Holes (Bond Model)

2-D representation of Si lattice:



C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 1-4

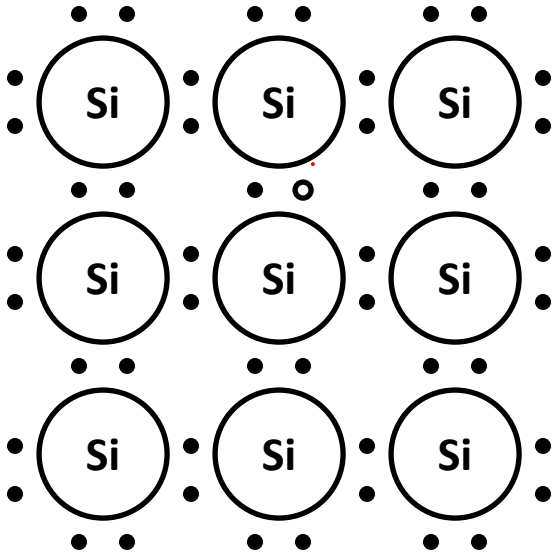
When an electron breaks loose and becomes a ***conduction electron***, a ***hole*** is also created.



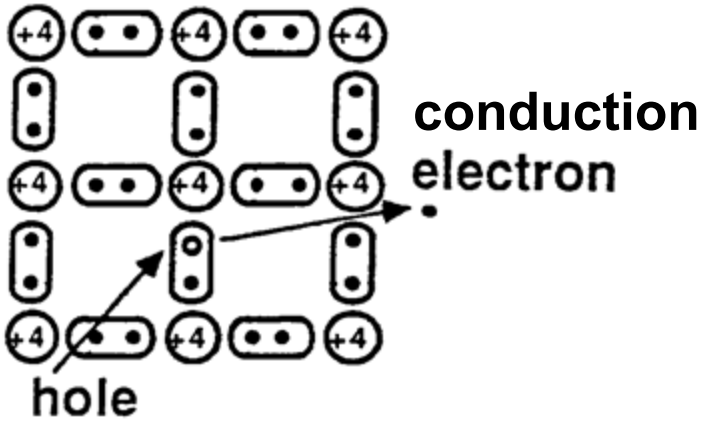
C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 1-5a

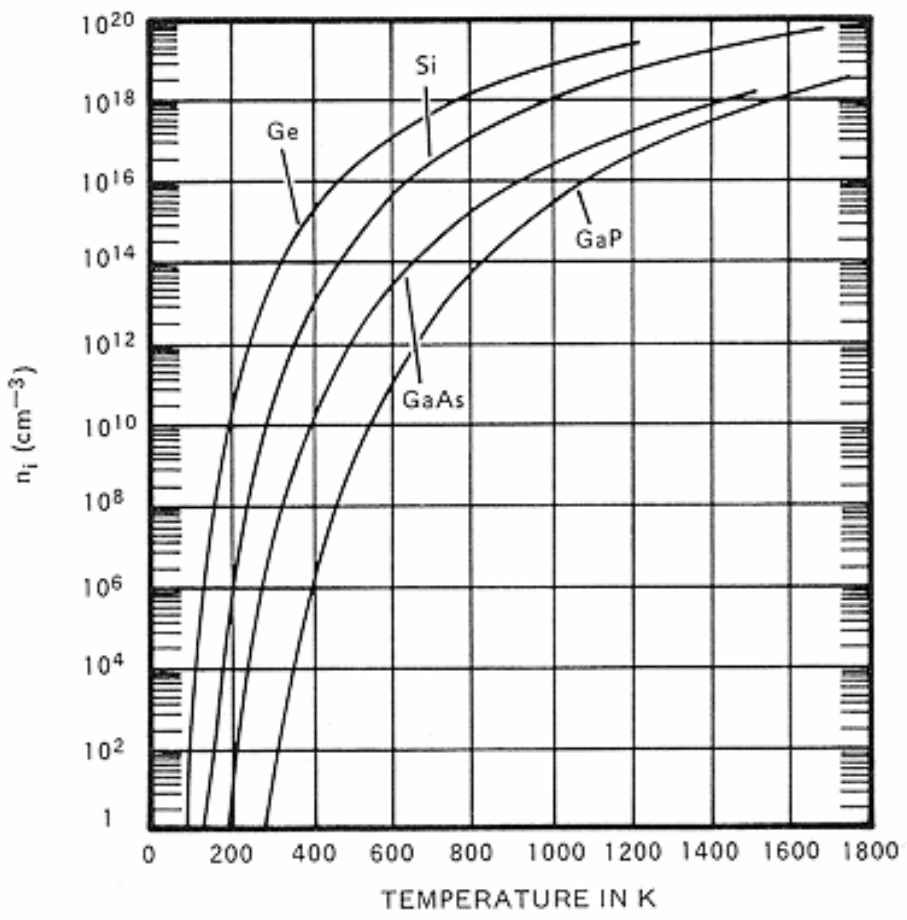
Recall: The Hole as a Positive Mobile Charge

- **Positive charge is associated with a half-filled covalent bond**
  - Moves when an electron from a neighboring covalent bond fills it



Recall: Intrinsic Carrier Concentration, n_i





- At temperatures > 0 K, some electrons will be freed from covalent bonds, resulting in electron-hole pairs.

For Si: $n_i \cong 10^{10}$ cm^{-3} at room temperature

Definition of Terms

$n \equiv$ number of electrons/cm^3

$p \equiv$ number of holes/cm^3

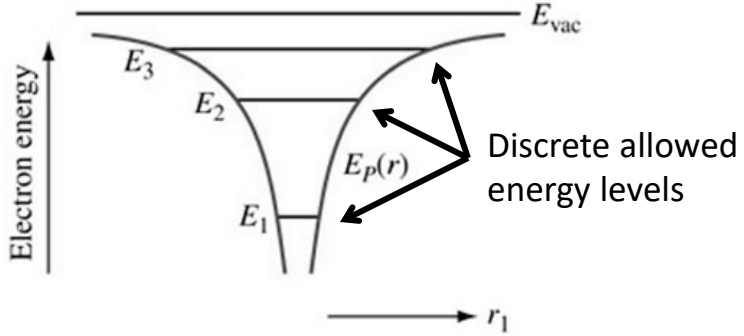
$n_i \equiv$ intrinsic carrier concentration

In a pure semiconductor,

$$n = p = n_i$$

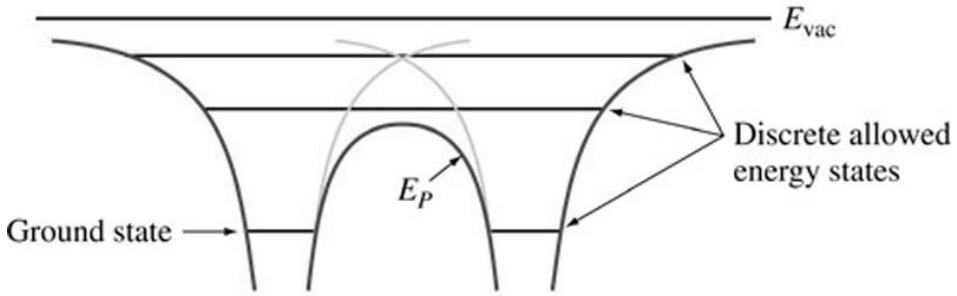
Potential Energy Profiles

1 atom



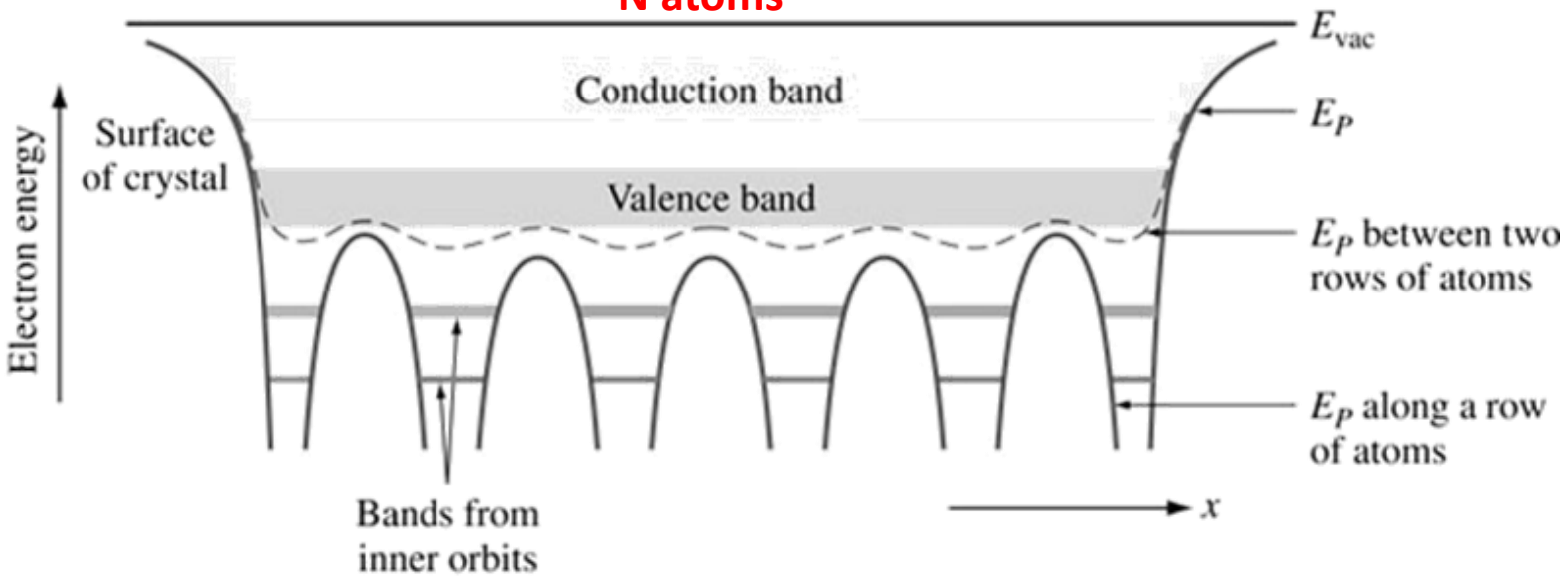
$V(r) \propto 1/r$ is mostly a coulombic potential btwn the positive nucleus & negative electrons.

2 atoms

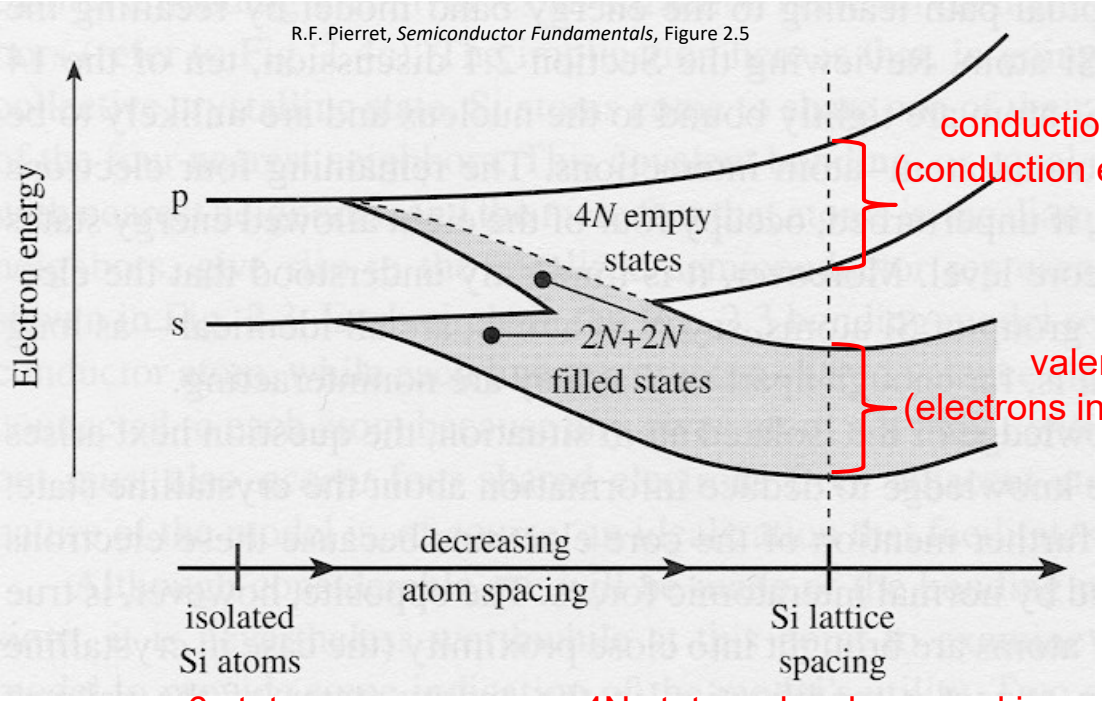


When two atoms are in close proximity, the upper energy levels are shifted to bonding and anti-bonding levels.

N atoms



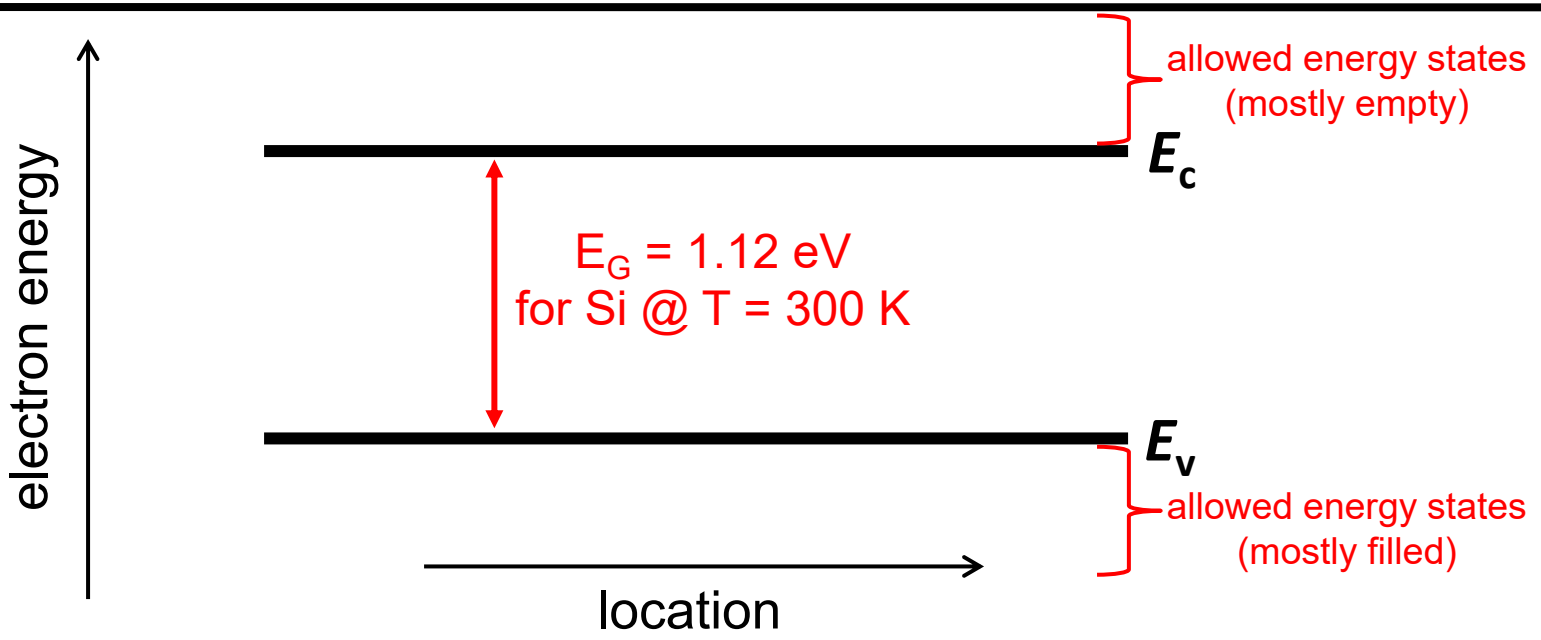
Si: From Atom to Crystal (N atoms)



Energy states in Si atom → energy bands in Si crystal

- The highest nearly-filled band is the ***valence band***
- The lowest nearly-empty band is the ***conduction band***

Energy Band Diagram

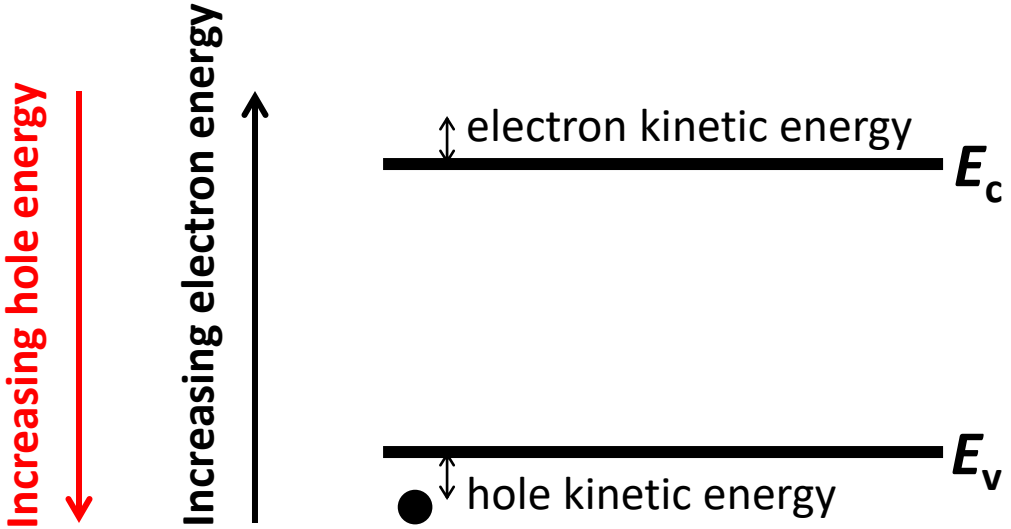


- Simplified version of energy band model, showing only the bottom edge of the conduction band ($\boldsymbol{E_c}$) and the top edge of the valence band ($\boldsymbol{E_v}$)
- E_c and E_v are separated by the **band gap energy $\boldsymbol{E_G}$**

Electrons and Holes (Band Model)

- Conduction electron = occupied state in the conduction band
- Hole = empty state in the valence band
- Electrons & holes tend to seek lowest-energy positions

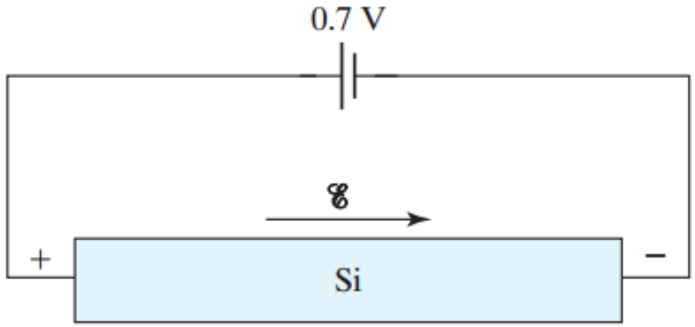
→ Electrons tend to fall and holes tend to float up (like bubbles in water)

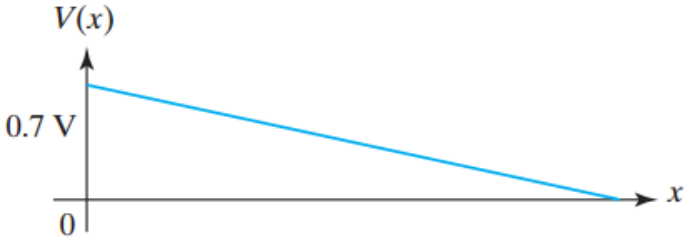


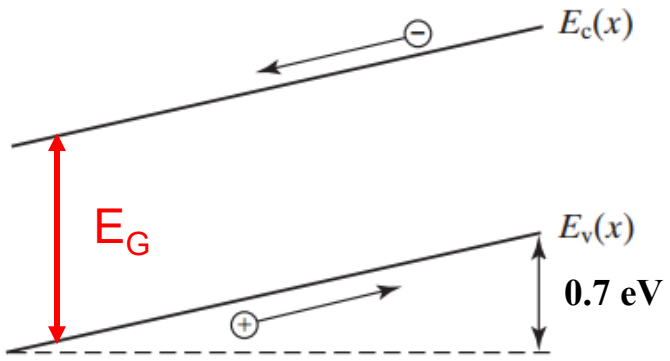
E_c represents the electron potential energy.

$$\text{P.E.} = E_c - E_{\text{reference}}$$

Electrostatic Potential, *V* and Electric Field, $\mathcal{E}$







- **The potential energy of a particle with charge *-q* is related to the electrostatic potential *V*(*x*):**

$$\text{P.E.} = -qV = Ec - E_{reference}$$

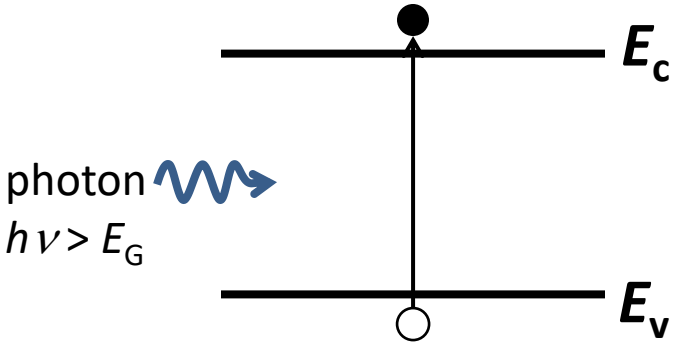
$$\Rightarrow V = \frac{1}{q}(E_{\text{reference}} - E_{\text{c}})$$

$$\boxed{\mathcal{E} = -\frac{dV}{dx} = \frac{1}{q}\frac{dE_{\text{c}}}{dx}}$$

- Variation of E_c with position is called ***"band bending."***

Measuring the Band Gap Energy

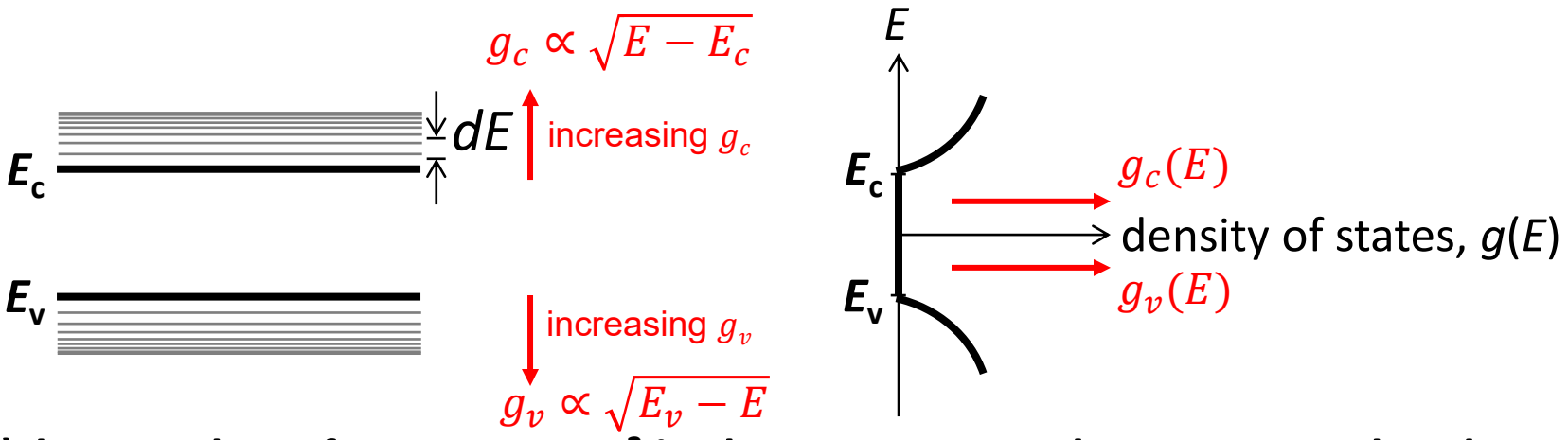
- E_G can be determined from the minimum energy of photons that are absorbed by the semiconductor



Band gap energies of selected semiconductors

Semiconductor	**Ge**	**Si**	**GaAs**
Band gap energy (eV)	0.67	**1.12**	1.42

Density of States



g*(*E*)*dE* = number of states per cm³ in the energy range between *E* and *E*+*dE

Near the band edges:

$$g_c(E) = \frac{8\pi\sqrt{2}}{h^3}\left(m^*_{n,DOS}\right)^{3/2}\sqrt{(E - E_c)} \quad \text{for } E \geq E_c$$

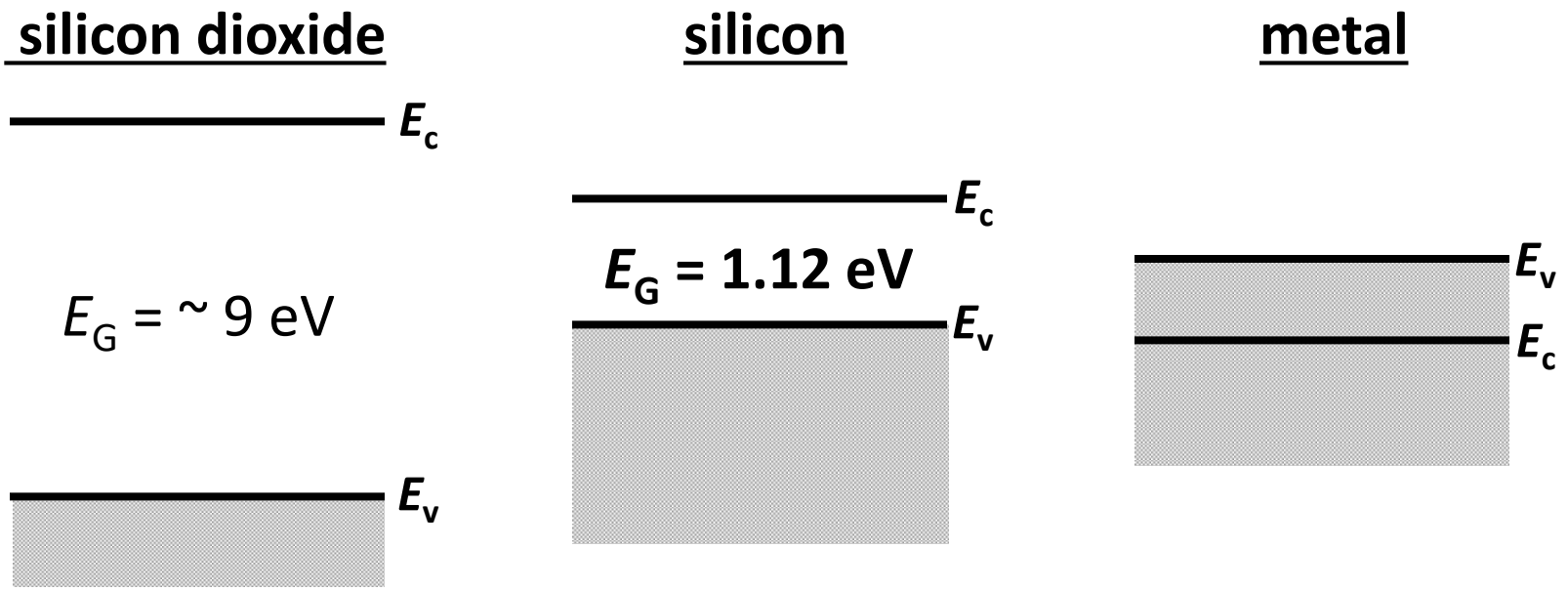
$$g_v(E) = \frac{8\pi\sqrt{2}}{h^3}\left(m^*_{p,DOS}\right)^{3/2}\sqrt{(E_v - E)} \quad \text{for } E \leq E_v$$

Electron and hole density-of-states effective masses

	Si	**Ge**	**GaAs**
$m_{n,DOS}^*/m_o$	**1.08**	0.56	0.067
$m_{p,DOS}^*/m_o$	**0.81**	0.29	0.47

$m_0 = 9.1 \times 10^{-31}$ kg

E_G and Material Classification



- Neither filled bands nor empty bands allow current flow
- Insulators have large E_G
- Semiconductors have small E_G
- Metals have no band gap (conduction band is partially filled)

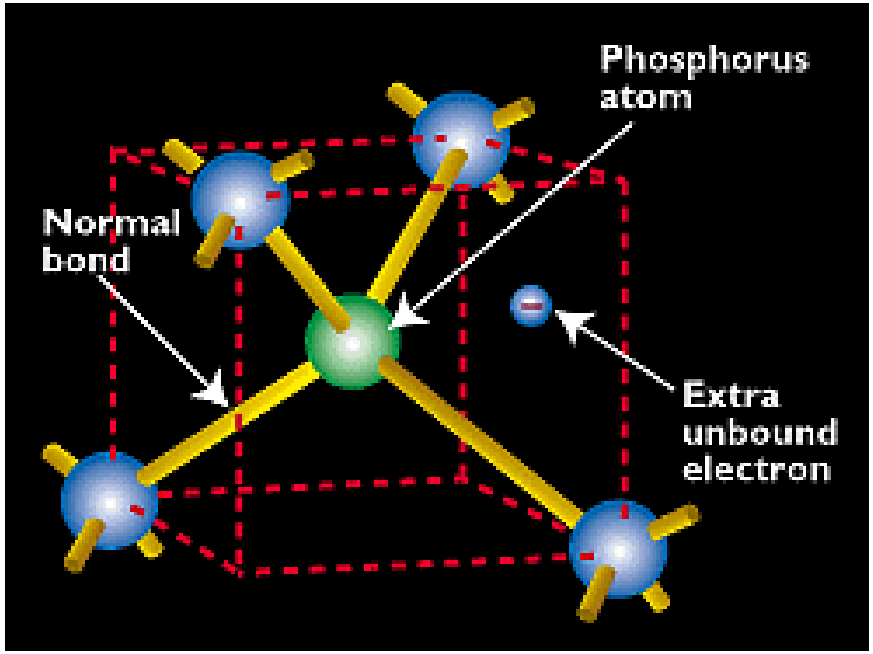
Doping

Typically, 10^{15} to 10^{20} cm^{-3}

- By substituting a Si atom with a special impurity atom (Column V or Column III element), a conduction electron or hole is created.

Column V

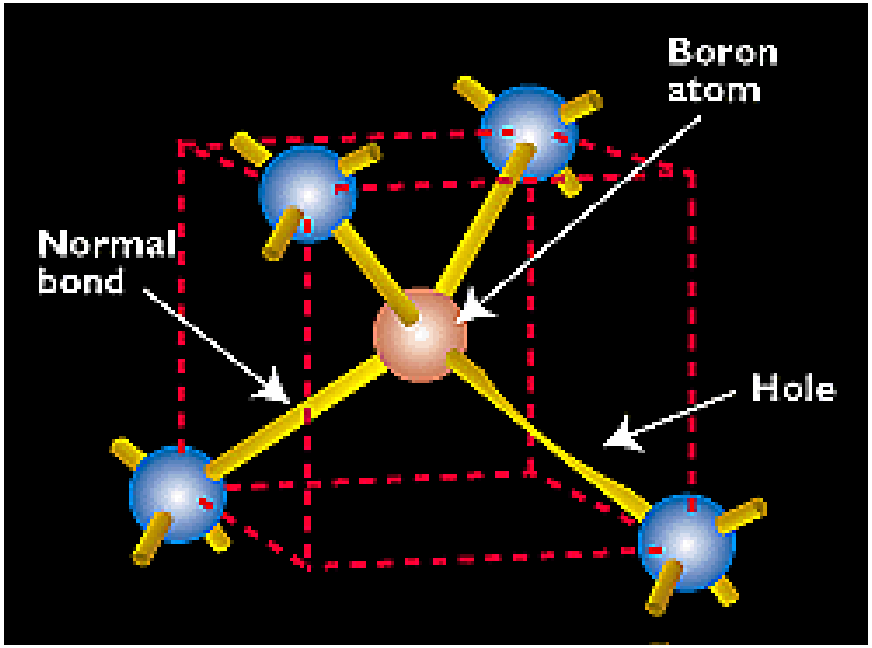
Donors: P, As, Sb



N_D ≡ ionized donor concentration (cm^{-3})

Column III

Acceptors: B, Al, Ga, **In**

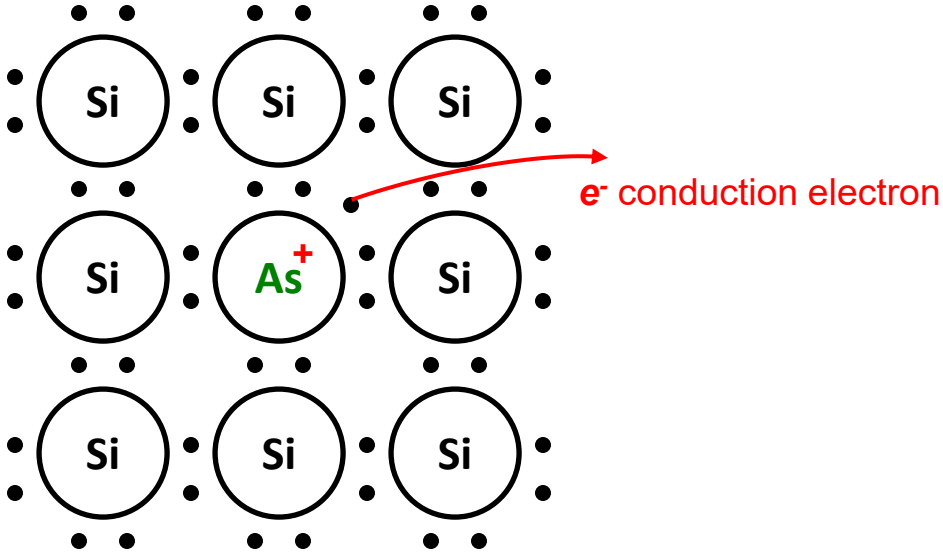


N_A ≡ ionized acceptor concentration (cm^{-3})

http://inventors.about.com/library/inventors/blsolar5.htm

Doping Silicon with a Donor

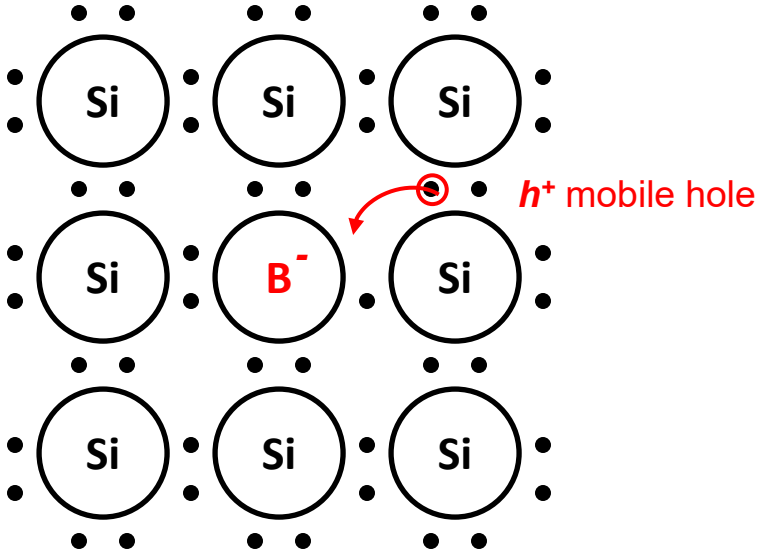
Example: Add arsenic (As) atom to the Si crystal



The loosely bound 5th valence electron of the As atom “breaks free” and becomes a mobile electron for current conduction.

Doping Silicon with an Acceptor

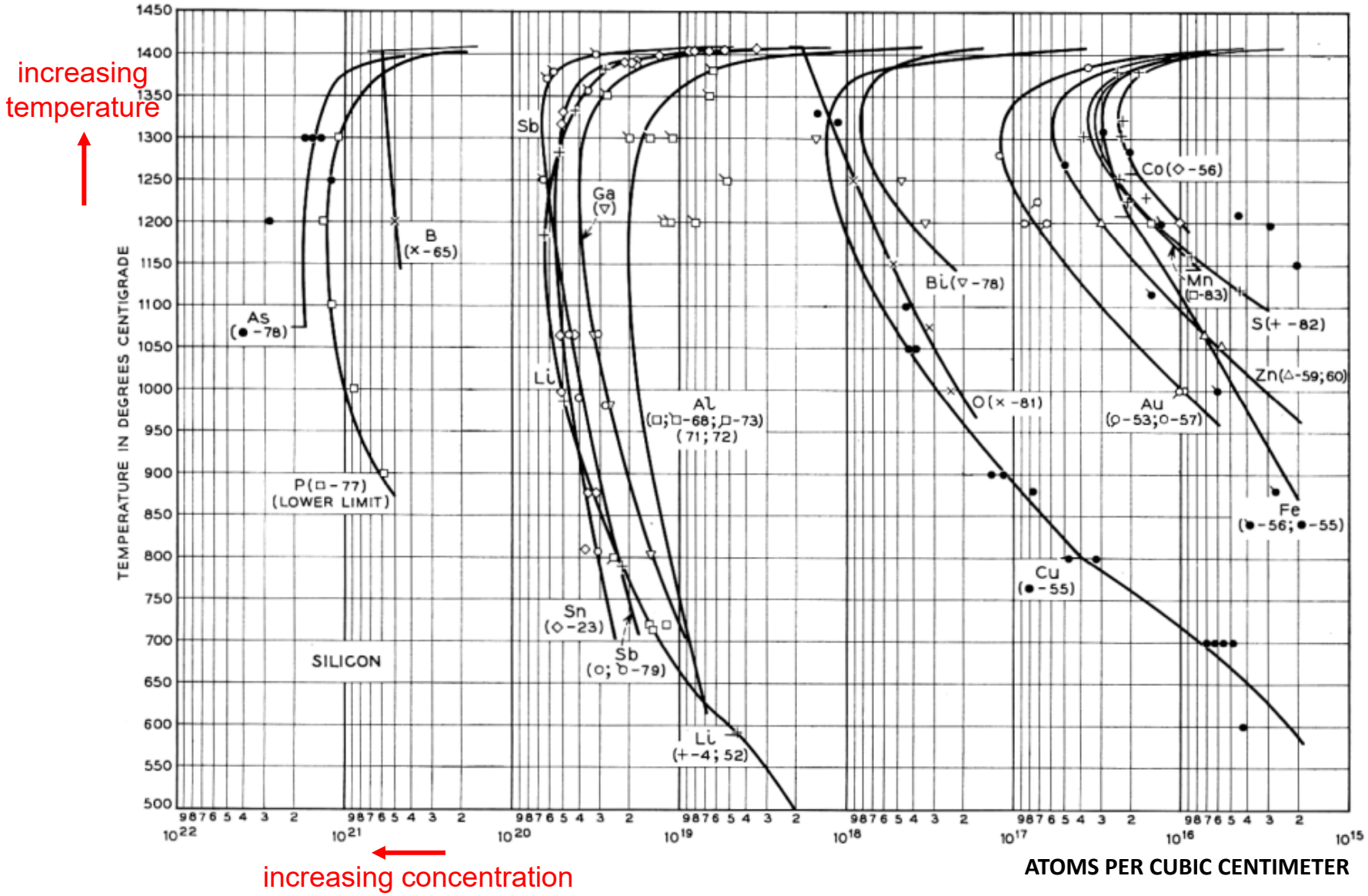
Example: Add boron (B) atom to the Si crystal



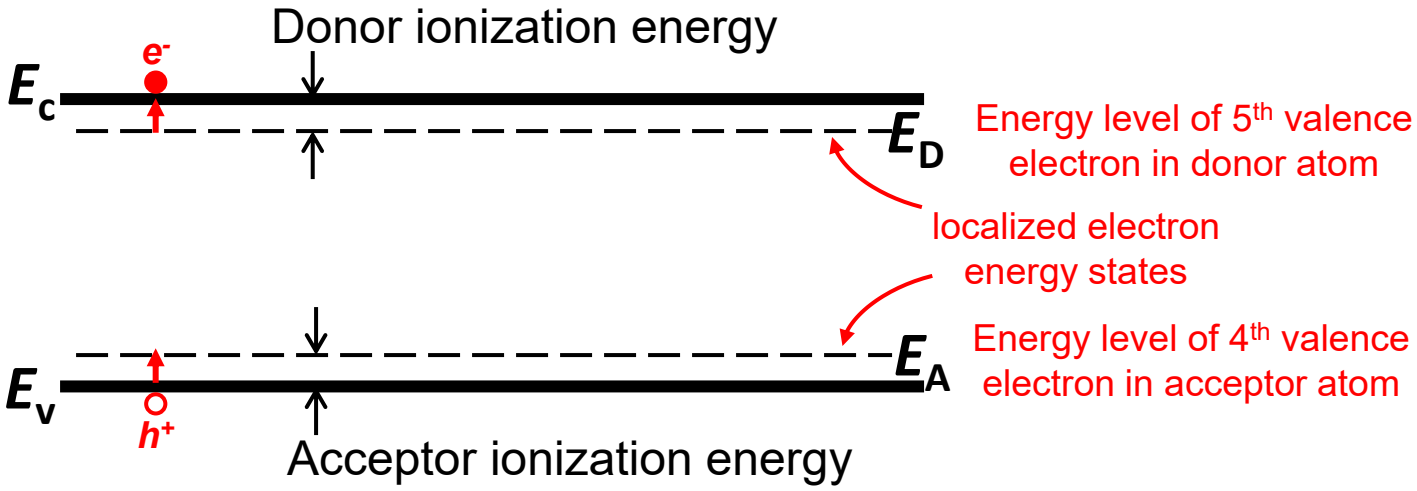
The B atom accepts an electron from a neighboring Si atom, resulting in a missing bonding electron, or “hole”. The hole is free to roam around the Si lattice, carrying current as a positive charge.

Solid Solubility of Dopants in Si

F. A. Trumbore, Bell Systems Technical Journal, 1960



Doping (Band Model)

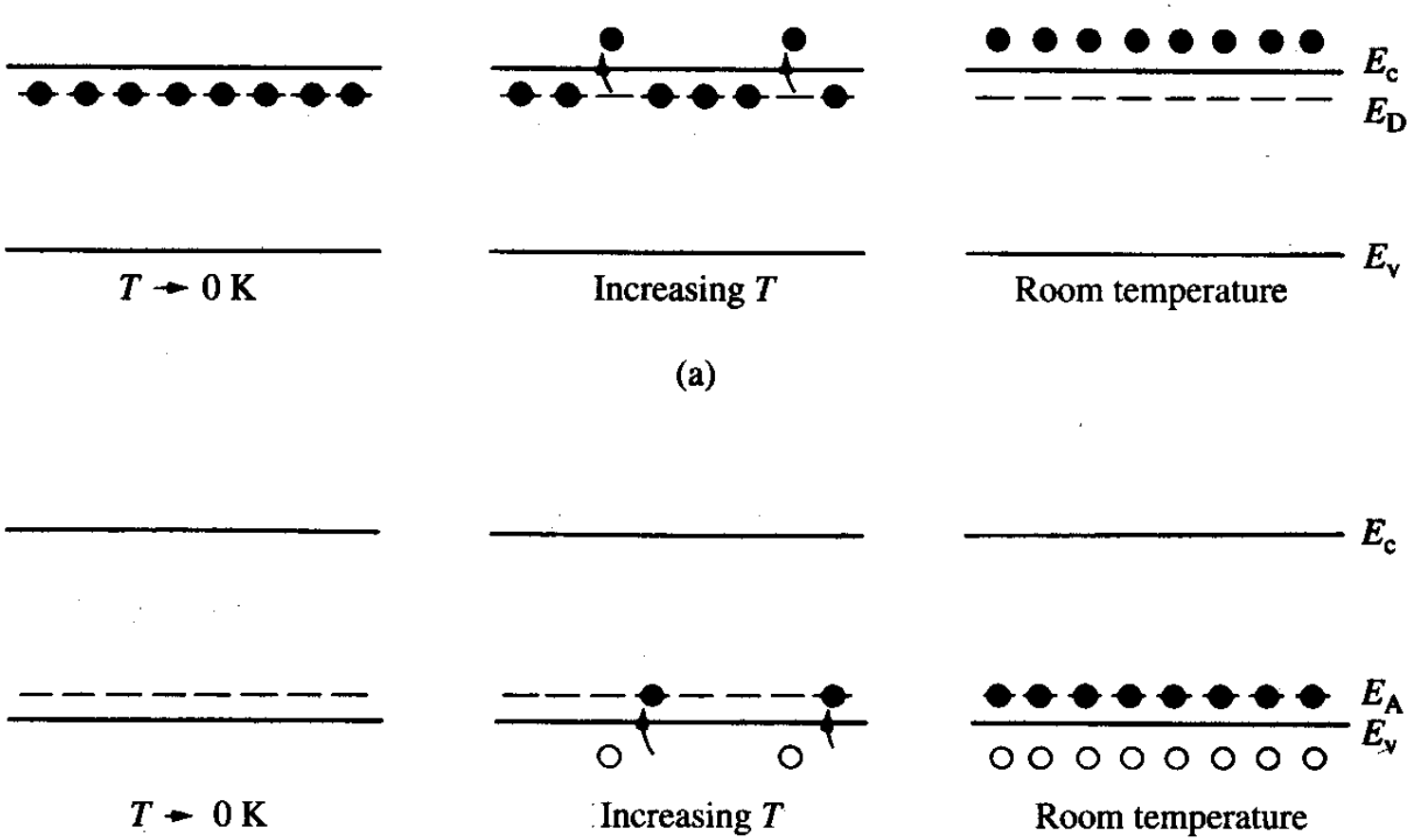


Ionization energy of selected donors and acceptors in silicon

	Donors			**Acceptors**		
Dopant	Sb	**P**	**As**	**B**	Al	In
Ionization energy (meV) E_c-E_D or E_A-E_v	39	**45**	**54**	**45**	67	160

Dopant Ionization

R.F. Pierret, *Semiconductor Fundamentals*, Figure 2.13



Charge-Carrier Concentrations

negatively charged

Charge neutrality condition: $N_D + p = N_A + n$

positively charged

At thermal equilibrium, $np = n_i^2$ **("Law of Mass Action")**

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

Note: Carrier concentrations depend on *net* dopant concentration!

n-type Material ($n > p$)

$N_D > N_A$ (more specifically, $N_D - N_A >> n_i$)**:**

Example: $N_D = 10^{17}$ cm^{-3} $N_A = 0$

negligible

$$n = \left(\frac{N_D - N_A}{2}\right) + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + {n_i}^2} = \left(\frac{10^{17}}{2}\right) + \sqrt{\left(\frac{10^{17}}{2}\right)^2 + (10^{10})^2}$$

$$\approx \left(\frac{10^{17}}{2}\right) + \sqrt{\left(\frac{10^{17}}{2}\right)^2} = 10^{16}\ \text{cm}^{-3}$$

$$p = \frac{{n_i}^2}{n} = \frac{(10^{10})^2}{10^{17}} = 10^3\ \text{cm}^{-3}$$

Note $n \gg p$!

p-type Material ($p > n$)

$N_A > N_D$ (more specifically, $N_A - N_D >> n_i$)**:**

Example: $N_A = 10^{16}$ cm^{-3} $N_D = 0$

negligible

$$p = \left(\frac{N_A - N_D}{2}\right) + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + {n_i}^2} = \left(\frac{10^{16}}{2}\right) + \sqrt{\left(\frac{10^{16}}{2}\right)^2 + (10^{10})^2}$$

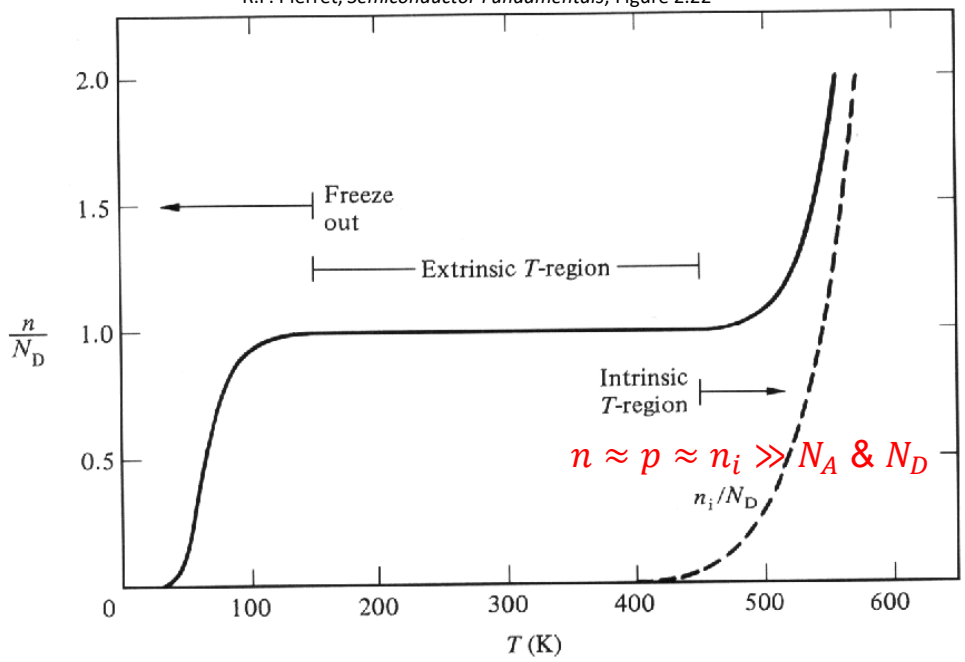
$$\approx \left(\frac{10^{16}}{2}\right) + \sqrt{\left(\frac{10^{16}}{2}\right)^2} = 10^{16}\ \text{cm}^{-3}$$

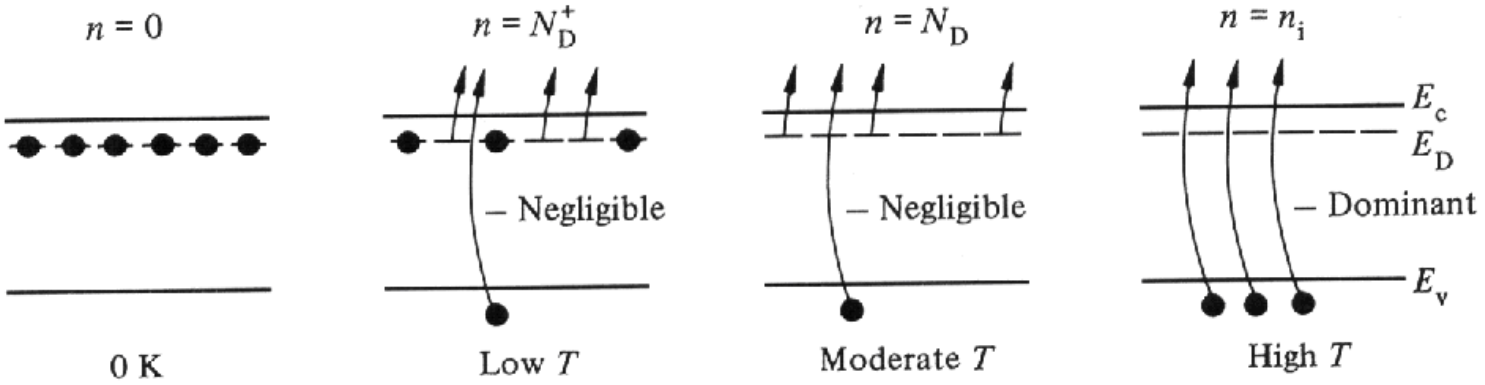
$$n = \frac{{n_i}^2}{p} = \frac{(10^{10})^2}{10^{16}} = 10^4\ \text{cm}^{-3}$$

Note $p \gg n$!

Carrier Concentration *vs.* Temperature

R.F. Pierret, *Semiconductor Fundamentals*, Figure 2.22





Terminology

donor: impurity atom that increases *n*

acceptor: impurity atom that increases *p*

n-type material: contains more electrons than holes

p-type material: contains more holes than electrons

majority carrier: the most abundant carrier

minority carrier: the least abundant carrier

intrinsic semiconductor: $n = p = n_i$

extrinsic semiconductor: doped semiconductor

such that majority carrier concentration = net dopant concentration

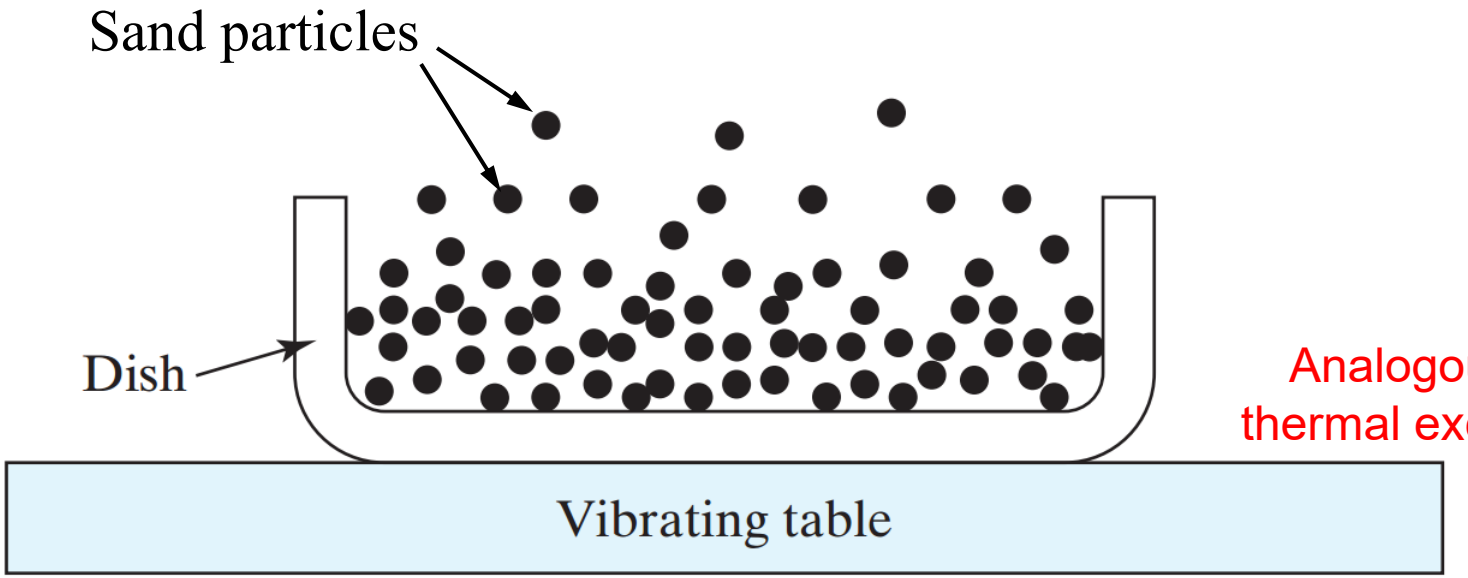
Thermal Equilibrium

- No external forces are applied:
  - electric field = 0, magnetic field = 0
  - mechanical stress = 0
  - no light
- Dynamic situation in which every process is balanced by its inverse process

  Electron-hole pair (EHP) generation rate = EHP recombination rate
- Thermal agitation → electrons and holes exchange energy with the crystal lattice and each other

  → Every energy state in the conduction band and valence band has a certain probability of being occupied by an electron

Analogy for Thermal Equilibrium



Analogous to thermal excitation

C. C. Hu, *Modern Semiconductor Devices for Integrated Circuits*, Figure 1-17

- There is a certain probability for the electrons in the conduction band to occupy high-energy states under the agitation of thermal energy (vibrating atoms).

Fermi Function

- Probability that an available state at energy *E* is occupied:

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

- E_F is called the ***Fermi energy*** or the ***Fermi level***

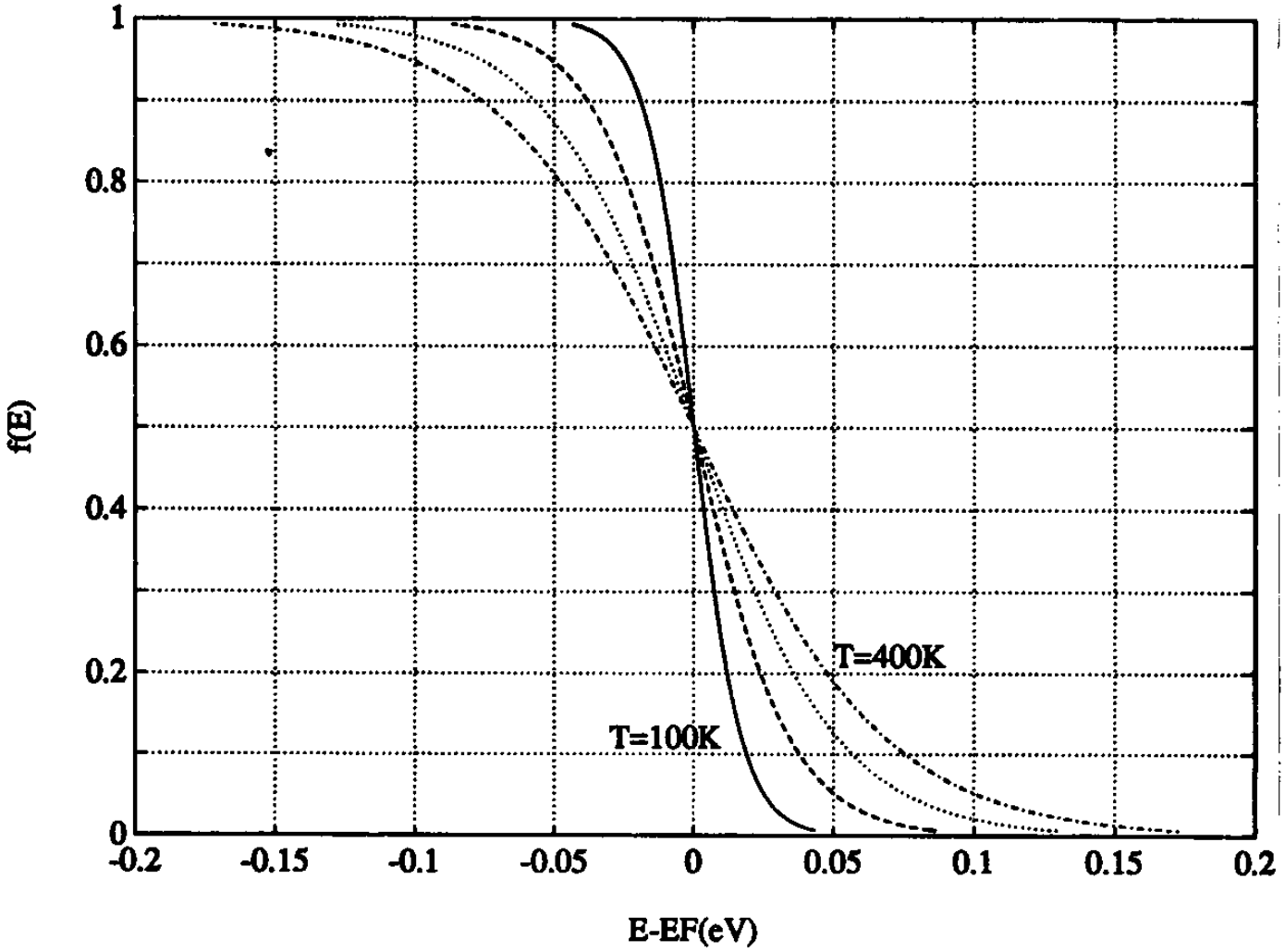
There is only one Fermi level in a system at equilibrium.

If $E >> E_F$: $f(E)$ approaches 0

If $E << E_F$: $f(E)$ approaches 1

If $E = E_F$: $f(E) = \frac{1}{2}$

Effect of Temperature on *f*(*E*)



Boltzmann Approximation

$$\text{If } E - E_F > 3kT,\ f(E) \cong e^{-(E-E_F)/kT}$$

$$\text{If } E_F - E > 3kT,\ f(E) \cong 1 - e^{(E-E_F)/kT}$$

Probability that a state is empty (*i.e.* occupied by a hole):

$$1 - f(E) \cong e^{(E-E_F)/kT} = e^{-(E_F-E)/kT}$$

Summary

- Allowed electron energy levels in an atom give rise to bands of allowed electron energy levels in a crystal.
  - The valence band is the highest nearly-filled band.
  - The conduction band is the lowest nearly-empty band.
- The band gap energy is the energy required to free an electron from a covalent bond.
  - **E_G for Si at 300 K = 1.12 eV**
  - Insulators have large E_G; semiconductors have small E_G

Summary (cont'd)

- **E_c represents the electron potential energy**

  Variation in $E_c(x)$ → variation in electric potential V

  Electric field $\mathcal{E} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_v}{dx}$

- **E - E_c represents the electron kinetic energy**

Summary (cont'd)

- **Dopants in silicon:**
  - Reside on lattice sites (substituting for Si)
  - Have relatively low ionization energies (<50 meV)
    → ionized at room temperature
  - **Group-V elements** contribute conduction electrons, and are called ***donors***
  - **Group-III elements** contribute holes, and are called ***acceptors***

Dopant concentrations typically range from 10^{15} cm^{-3} to 10^{20} cm^{-3}

Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes